FERT Report – Design, Calculations and Allocation Logic

Technical documentation for the Flask + PyMySQL program that builds `fert_report.xlsx` from the three SAP exports ZPLAN, ZMB52 (stock) and ZBOMEXPLODE. Every rule below is taken from the code (`columns.py` , `db.py` , `engine.py` , `report.py` , `app.py`). Where a rule was inferred from your sample `fert_report.xlsx` rather than from a written formula, it is marked **[inferred]**. Section 10 lists what is verified and what is not.

1. End-to-end flow

1. **Upload** ZPLAN, ZMB52 and ZBOMEXPLODE, and choose Plant (e.g. AC02), Shop (e.g. SH-3 or ALL), Purchase group (ALL or a code), Report date, and the dependency-matrix tick-box.
2. **Read** each file by column *position* (the SAP ".xls" files are really .xlsx; the app detects this).
3. **Load** into MySQL tables `zplan` , `zmb_stock` , `zbomexplode` (rows of that plant are deleted first, then re-inserted).
4. **Read back** the plant's rows from MySQL and run the allocation engine: FG stock → sub-assembly (SA) stock → brought-out component stock.
5. **Write back** allocation results into `zplan` (FG columns) and `zmb_stock` (assigned / balance columns).
6. **Build** the nine report sheets, filtered by Shop / Purchase group, and offer the download.

2. Input tables and the columns that are used

The Excel files and your MySQL tables have the same column order, so columns are mapped by position to clean internal names (`columns.py`). Column counts: ZPLAN 34, ZMB52 30, ZBOMEXPLODE 62.

2.1 ZPLAN (`zplan`) – the production / spare plan

Column	Used for
Plant, Shop	filters

Column	Used for
Material	the FERT or SPARE material (plan line key together with Plan_week and Priority)
Plan_week, Plan_month	ordering of allocation; month columns of the Summary
Priority	allocation order inside a week (shared by all shops)
Plan_Qty	quantity to be produced / supplied
Plan_Value	value shown in PLAN and Summary
MM_Price	price used to value FG stock assigned
Sales_Document, Item_SD	sales-order specific stock (if any)
Fert_or_Spare_Plan	splits PLAN sheet into FERT and SPARE sections and Summary rows
Product, Material_Description, Requested_deliv_date	PLAN columns
Original_FG_Stock, Assigned_FG_Qty, FG_Stock_after_Assignment, Balance_Plan_Qty_to_be_Produce, FG_Assigned_Value	**calculated by the program** and written back

2.2 ZMB52 (`zmb_stock`) – plant stock

Column	Used for
Plant, Material, Material_Type, Material_Description, Material_Group, Base_Unit_of_Measure	USABLE & UNUSABLE sheet
Total_original_Stock_Qty	the stock pool that is allocated (consolidated stock)
Total_Original_Stock val	stock value; unit value = value ÷ quantity
Vendor_Code	rows with a vendor code are vendor stock and are **not** part of plant stock
Sales_Document, Item_SD	sales-order stock pools

Column	Used for
ABC_Indicator, Procurement_type, Special_procurement	USABLE sheet; Procurement_type = F/E drives Summary
Purchasing_Group, Description_p_group	purchasing group name used in every sheet ("Pur.Name")
Assigned_Stock_Qty, Assigned_Stock_Value, Balance_Stock_Qty, Balance_Original_Stock_val	**calculated by the program** and written back

2.3 ZBOMEXPLODE (`zbomexplode`) – exploded BOM per plan line

Column	Used for
Plan_week, Priority, Fert, Sales_Document, Item_SD	identify the plan line (one explosion per line)
Running_Serial_number, Level	rebuild the BOM tree (explosion order)
Component, Material_Description, Material_Group, Base_Unit_of_Measure	component identity
Parent_Code, Parent_Material_Desc	parent assembly of the row
Parent_Code_Proc_Type, Parent_Code_Spl_Proc_Type	parent procurement type (F = brought-out, E = in-house); special procurement 30 = sub-contract, 50 = phantom assembly
Comp_Code_Proc_Type, Comp_Code_Spl_Proc_Type	component procurement type (F → Brought Out sheet, otherwise In house sheet)
Phantom_item	phantom rows are never reported
Total_Org_BomQty, Total_Org_BomVal	original requirement quantity / value ("Req.Qty / Req.Val as per BOM")
Total_Bom_Qty	requirement of the node (used for "SA Required Qty")
Bom_Qty, SA_Qty, Plan_Qty	per-unit BOM qty, SA qty per FERT, plan qty
Unit_price, Price_source, Latest_Inforecord_Price	price, price source, latest info-record price
Dependency_Matrix_Code_1 / _2	SA codes that a component depends on

Column	Used for
Purchasing_Group, ABC_Indicator, Minimum_Lot_Size, Reorder_Point	purchase group, ABC class, MOQ sheet
Shop, Product, Fert_Description, Plan_month, Requested_deliv.date	filters / descriptive columns

2.4 Lookups that are **not** in the three tables

- Material-group description ("Mat.Grp.Desc") and Product description come from `lookups/mat_group_desc.csv` and `lookups/product_desc.csv`, extracted from your sample report. Add new groups/products to those CSVs.
- "Routing Group" (In-house sheet) is not available in the three tables and is left blank.

3. Data preparation (load step)

- Blank cells become NULL; text is trimmed; numbers become `float` / `int`; dates are parsed.
- Each value is checked against the MySQL column type. If a decimal column is too small for the data (scale or range) or a varchar is too short, the load stops with the exact `ALTER` statements (or widens automatically when `AUTO_WIDEN_COLUMNS=1`).
- **Important:** your original schema has `decimal(x,2)` columns that would round 3-decimal quantities (`Bom_Qty`, `Total_Bom_Qty`, `SA_Qty`, `Total_Org_BomQty`, and three stock quantity columns). Run `schema_fix.sql` once.
- Loading **replaces** the plant's rows in each table, so re-uploading the same plant never duplicates data.

4. Stock pools

- A pool key is `(Material, Sales_Document, Item_SD)`. Stock without a sales document goes to the plant pool `(Material, '', '')`.
- Pool quantity = sum of `Total_original_Stock_Qty` (consolidated stock).
- Vendor stock rows (Vendor_Code filled) are ignored.
- A plan line / BOM row with a sales document first looks for its own sales-order pool; if none exists it uses the plant pool.
- All stages draw from the **same** pools in sequence, so stock used by FG is no longer available for SA or components.

5. How stock is assigned

5.1 Allocation order

Plan_week ascending (text `YYYY-WW`), then Priority ascending (numeric), then original file order. All shops are allocated together because stock is plant-wide; the shop filter is applied only when the report is written.

5.2 Stage 1 – FG (finished goods / spare) stock, per ZPLAN line

```
Original_FG_Stock          = pool quantity of the Material before this line
Assigned_FG_Qty            = MIN(Plan_Qty, Original_FG_Stock)
FG_Stock_after_Assignment  = Original_FG_Stock − Assigned_FG_Qty
Balance_Plan_Qty           = Plan_Qty − Assigned_FG_Qty
FG_Assigned_Value          = Assigned_FG_Qty × MM_Price
FG ratio of the line       = Balance_Plan_Qty ÷ Plan_Qty
```

The FG ratio scales the BOM requirement: if 4 of 5 pieces come from FG stock, only 1/5 of the BOM is needed.

5.3 Stage 2 – rebuild the BOM tree

For each plan line (group = week | priority | fert | sales doc | item) rows are sorted by Running_Serial_number. A row's **parent is the most recent row with Level − 1**. A row that is the parent of another row "has children" (it is a sub-assembly node). This reconstruction matched the Parent_Code column on 4,081 of 4,081 rows in your data.

5.4 Stage 3 – requirement and stock for each row

Definitions per row: `Org` = Total_Org_BomQty; `ratio` = FG ratio of its plan line; `pf` = coverage factor of its parent (1 for first-level rows).

```
R (requirement after FG and SA stock) = Org × ratio × pf
```

Rows that are skipped / excluded

- Phantom rows (Phantom_item filled): never allocated, never reported; they pass `pf` through to their children.
- **Brought-out parent rule:** a row is *excluded* (not reported, requirement 0) when its parent is procurement type F **and** the parent's special procurement is not 30, or when its parent is already excluded. Reason: nothing below a bought-out part needs to be purchased. Parents with special procurement 30 (sub-contract) are different: their components are still reported, because they are supplied to the vendor.
- Text-only BOM lines (no Component) are kept for the In-house sheet with zero values.

Phase A – sub-assembly rows (rows that have children), parents before children

```
if Comp_Proc_Type is F or E, Component exists and R > 0:
    Assigned = MIN(R, pool of Component)      (pool reduced)
    coverage factor of the node = pf × (R − Assigned) ÷ R
else: factor = pf (or 0 when R = 0)
```

The coverage factor is what reduces the children. Example: an SA needs 5, 3 are available in stock → factor 0.4, so every component below it is required only 40 %.

Phase B – leaf components (no children), brought-out (F) only

```
allowed  = R × dependency fraction           (1 when the rule is off or not
applicable)
Assigned = MIN(allowed, pool of Component)
```

In-house (E) leaf components are reported but **not** allocated from stock (only brought-out items are, as in the SAP logic).

Dependency matrix (optional tick-box). A component may name up to two SA codes (Dependency_Matrix_Code_1/_2). For each code that (a) has plant stock and (b) exists as an SA node in the same plan line, the fraction is `Assigned ÷ R` of that SA; the component may take at most `R × MIN(fractions)`. Codes without stock, or not found in the line, are ignored. Example: components of a stator needing 5 SAs, SA stock gave 3 → at most 3 of 5 may be allocated. **[inferred]** – see section 10.

5.5 Derived row values

```
Shortage Qty            = R − Assigned               (0 if ⩽ 0)
Req.Qty as per MRP      = R if Shortage > 0 else 0
Price                   = Unit_price
Req.Val as per BOM      = Total_Org_BomVal
Req.Val as per MRP      = Req.Qty as per MRP × Price
Assigned Value          = Assigned × Price
Shortage Value          = Shortage × Price
SA Required Qty         = Total_Bom_Qty of the parent row × ratio
                          (first level: SA_Qty × Plan_Qty × ratio)
NO InfoRecord           = "NO-PRICE-LINK" if Price_source = MM_PRICE
                          "MISSING-VIEW"  if Price_source is blank
```

5.6 Stock write-back (`zmb_stock`)

```
Assigned_Stock_Qty = original pool − remaining pool
Balance_Stock_Qty  = remaining pool
Assigned/Balance value = quantity × (Total_Original_Stock val ÷
Total_original_Stock_Qty)
```

Rows sharing one pool split the result pro-rata to their quantity.

6. The nine sheets

All sheets start with a title row ("… as on date -dd/mm/yyyy"); most have a second line ("Purchase group: …" / "Plan for: ALL"), a blue header row, and a running Srl. number. Quantities use 2–3 decimals as in your report.

6.1 MISSING VIEW

Unique components (plant-wide, all shops) from the brought-out BOM rows whose Price_source is blank (no MRP/purchase view). Columns: Srl., Plant, Component, Component Desc, Material Group, Mat.Grp.Desc, Pur.group, Name (purchasing group name). Sorted by component.

6.2 PLAN

ZPLAN lines of the plant and shop, sorted by Plan_week then Priority. Two blocks: **FERT** then **SPARE** (by Fert_or_Spare_Plan), each with its own Srl. numbering. Columns: Plant, Product, Product Desc (lookup), FERT/SPARE code, description, Plan Week, Plan Month, Del Date, Priority (4 digits, e.g. 0037), Plan Qty, Plan Val, Assigned Qty = Assigned_FG_Qty, Assigned Value = FG_Assigned_Value, Remark (blank).

6.3 STOCK ALLOCATION-Brought Out

BOM rows where Comp_Proc_Type = F, component present, not phantom, not excluded, filtered by shop and purchase group. Sorted by Plan_week, Priority, parent level, Parent Code, Component. Columns (35): product, FERT, plan data, Parent Code/Desc, SA Required Qty, Component data, Price, purchasing group/name, procurement types, ABC, Req.Qty/Val as per BOM, Req.Qty/Val as per MRP, Assigned Qty/Value, Shortage Qty/Value, Price Source, Latest Price, NO InfoRecord. Formulas are in 5.5.

6.4 STOCK ALLOCATION- In house

Same selection, but Comp_Proc_Type is not F (E, X, and text-only lines). Same columns up to Shortage Value, then Routing Group (blank). Assigned Qty is non-zero only for in-house sub-assembly nodes that have stock (Phase A).

6.5 SHORTAGE LIST

Brought-out rows with Shortage Qty > 0. Columns as above with "Required Qty/Val" = Req as per BOM, then Assigned, Shortage, Price Source, NO InfoRecord.

6.6 USABLE & UNUSABLE STOCK

One row per ZMB52 row of the plant.

```
Usable Qty    = sum of Assigned Qty of the material in the Brought-Out and
In-house
                sheets (selected shop) + Assigned_FG_Qty of its PLAN lines
Usable Value  = same sum in value
Unusable Qty  = Total Qty − Usable Qty        Unusable Value = Total Val −
Usable Value
```

If a material has several stock rows the usable amount is shown on its first row only. Total Qty/Val = Total_original_Stock_Qty / Total_Original_Stock val.

6.7 MOQ VAL

Brought-out components (plant-wide) with Minimum_Lot_Size > 0, one row per component.

```
Shortage Qty as per BOM = sum of shortage over all lines
Free stock              = MAX(Total stock − sum of assigned, 0)
Net need                = Shortage + MAX(Reorder_Point − Free stock, 0)
As Per MRP Qty          = CEILING(Net need ÷ MOQ) × MOQ   (0 if Net need ⩽ 0)
Addtional MOQ Qty       = MAX(As Per MRP Qty − Shortage, 0)
Values                  = quantity × Unit Price
```

[approximation] SAP's own MRP lot-sizing result is not in the three tables.

6.8 NO PRICE LINK

Brought-out components with Price_source = MM_PRICE (no info record), grouped by component: Tot BOM Qty = sum of Req.Qty as per BOM, Tot PR Qty = sum of Req.Qty as per MRP; only components with Tot PR Qty > 0; INFOREC Price blank. **[inferred]**

6.9 Summary (values in lakhs = ÷ 100,000; month columns = distinct Plan Months of the shop's plan)

Row	Formula
Sales Plan Qty / Value	sum of Plan_Qty / Plan_Value of FERT plan lines, total and per month
Spare Plan Qty / Value	same for SPARE lines
Brought out – as per BOM	sum of Req.Val as per BOM (Brought-Out sheet), per month
Brought out – as per MRP	sum of Req.Val as per MRP, per month
Additional MOQ Value	sum of Addtional MOQ Val (MOQ VAL sheet), total only
Total Stock Value	sum of Total Val (USABLE sheet)
Total Value of Stock used	sum of Usable Value
– Procured (total)	Usable Value of stock rows with Procurement_type F; monthly cells = Assigned Value of Brought-Out rows per month
– Inhouse (total)	Total used − Procured used; monthly cells = Assigned Value of In-house rows
Total stock not used	Total stock − Total used
– Procured / Inhouse not used	Unusable Value of F stock / (Not used − Procured not used)
Total Shortage Value	sum of Shortage Value (Brought-Out), per month

Note: as in your report, the monthly cells of the Procured/Inhouse "used" rows add up to the allocation sheets only, while the TOTAL column also includes FG stock used.

7. Worked example (your data)

7.1 FG stage – spare A0222057810501, week 2026-44, priority 37, shop SH-3

Item	Value
Plan_Qty	5

Item	Value
Original_FG_Stock	4
Assigned_FG_Qty = MIN(5, 4)	4
FG_Stock_after_Assignment	0
Balance_Plan_Qty_to_be_Produce	1
MM_Price	179.39
FG_Assigned_Value = 4 × 179.39	717.56
FG ratio	1 ÷ 5 = 0.2 (its BOM requirement is scaled to 20 %)

The program reproduces SAP's stored ZPLAN allocation on 377 of 378 lines. The exception is spare A0222006060000 (week 2026-44, priority 38, plan 6, stock 4): the program assigns 4, but the ZPLAN you supplied shows 0.

7.2 BOM stage – FERT VB037IF3000, week 2026-44, priority 1, plan 5, FG stock 0 (ratio 1)

Tree: VB037IF3000 → MVB037IF3000FA → MVB037IF3000GA → components. Three brought-out (F) components under MVB037IF3000GA, all with SA_Qty 1 and Bom_Qty 1:

`Req.Qty as per BOM = Bom_Qty 1 × SA_Qty 1 × Plan_Qty 5 = 5`, so `R = 5 × 1 × 1 = 5` for each.

Component	Price	Plant stock (ZMB52)	Report: Assigned	Shortage	MRP Qty	Values (Assigned / Shortage)
A1422098070100	14.60	42 (23 assigned plant-wide)	5	0	0	73.00 / 0
A0322205170000	323.44	7 (all 7 assigned)	2	3	5	646.88 / 970.32
A0322107140000	526.22	2 (all 2 assigned)	0	5	5	0 / 2,631.10

Reading the table: for A0322205170000, `Shortage = 5 − 2 = 3`, `MRP qty = 5` (because shortage > 0), `Assigned Value = 2 × 323.44 = 646.88`, `Shortage Value = 3 × 323.44 = 970.32`. Earlier plan lines (higher priority or earlier week) had already taken 5 of the 7 pieces.

One sub-assembly below it shows the exclusion rule: component A0322183080102 sits under A0322107140000, which is procurement type F without special procurement 30, so it is excluded and not reported.

What the program shows when run only on the three files you uploaded: 3 / 3 / 2 for the three components. Two reasons: (1) your ZBOMEXPLODE contains only week 2026-44, so stock used by earlier weeks is not consumed first; (2) the dependency rule limits the first two components because their dependent SA H03294140AY1DR1I received only 3 of 5 from stock, while your report did not apply that limit. This is a good reason to compare runs with the tick-box on and off once you have the full BOM export.

8. Scenarios and how they are handled

Scenario	Behaviour
Material has no stock record	pool = 0 → Assigned 0, Shortage = R
FG stock fully covers the plan	ratio = 0 → R = 0 for all BOM rows; rows stay in the sheets with zero quantities
FG stock covers part of the plan	BOM requirement scaled by Balance ÷ Plan
Sub-assembly stock covers part	children scaled by (R − Assigned) ÷ R
Parent is F with no special procurement	children excluded
Parent has special procurement 30	children reported
Phantom item	skipped, children keep the parent's factor
Text-only BOM line	listed in In-house sheet, all values 0
Same stock needed by several lines	served in week → priority order until the pool is empty
Stock with a sales document	used first by lines with the same document and item
Vendor stock	not part of plant stock
Price_source blank / MM_PRICE	flagged MISSING-VIEW / NO-PRICE-LINK
Shop = ALL	no shop filter; Purchase group = ALL
Re-upload of the same plant	existing rows replaced
Column too narrow in MySQL	load stops with ALTER statements, nothing partial is committed
Priority missing	treated as 0

9. Operating notes

- Setup: run `schema.sql`, then `schema_fix.sql`; set `DB_HOST`, `DB_USER`, `DB_PASSWORD`, `DB_NAME`; `python app.py`; open `http://localhost:5000`.
- The load, allocation write-back and report run in one request; a failure rolls the database back.
- Plan weeks sort as text, so keep the `YYYY-WW` format.

10. Verification status and known limitations (please read)

Checked against your sample `fert_report.xlsx` using the supplied files:

- PLAN: 309 rows, totals identical. In-house rows (328), stock rows (5,159) identical. Sales/Spare plan and Total Stock Value in Summary identical.
- FG allocation: 99.7 % of ZPLAN lines identical to the stored SAP values.
- Brought-out rows for week 2026-44: 1,991 of 1,991 report rows found, in the same order; the program lists 14 extra rows (insulation items `U2IM…` and two `A40CR…` items). The rule that excludes them in your report was not found.
- SA Required Qty: about 97.7 % identical for week 2026-44.
- Assigned/Shortage for week 2026-44, when stock already used by earlier weeks is taken into account: about 97.8 % identical.

Not verifiable with the supplied files:

- ZBOMEXPLODE holds only week 2026-44 (4,152 rows) while your report was built from a full BOM (about 23,500 brought-out rows). BOM-driven totals (Summary rows 10–22, shortage values) therefore differ until the full export is used.
- Dependency-matrix rule (5.4), NO PRICE LINK and MISSING VIEW inclusion rules, and the MOQ "As Per MRP" quantity are inferred or approximated.
- The formulas in `Zstock_allocation.docx` are largely screenshots; the logic here was reconstructed from the document text plus your sample report.
- MySQL load and write-back could not be executed in the build environment; type checks were tested against your table definitions.